The newsroom that speaks your language.

A multilingual, agent-run newsroom for the people local media reaches least. Built in Oulu, with a named human behind every published story.

SEVEN LANGUAGES

English

Suomi

Svenska

العربية

Davvisámegiella

Українська

Русский

FRONTEND / BACKEND / AGENTIC NEWSROOM

THE GAP

Local news misses the people who need it most.

01 **Newcomers in Oulu**

Permits, work, housing and language, explained in the language they read.

02 **Seasonal workers**

Contracts, staff housing and winter safety, without a translation delay.

03 **North Sámi speakers**

News in Davvisámegiella, reviewed by a person, not machine text.

IN THE PRODUCT

7

languages, every story

10 communities

22 topics, 11 links

The news exists. The language and the trust are missing.

WHAT WE BUILT

One story. Seven languages. One accountable newsroom.



AGENTS DO THE WORK

Monitor, cluster, investigate, write, translate, check.

PEOPLE HOLD THE GATE

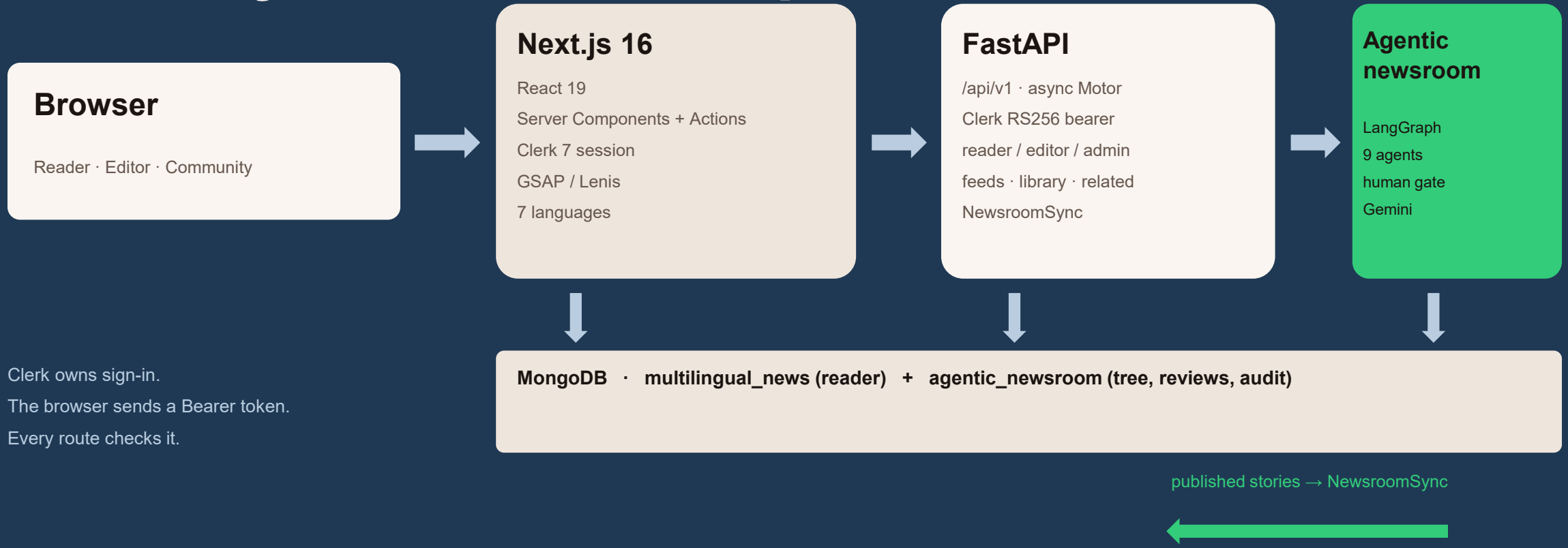
A named editor, language reviewer or subject expert approves.

READERS SEE WHY

Sources, reasons, corrections and the name behind every approval.

Nothing publishes without a named person.

Three systems, one loop.



Clerk owns sign-in.
The browser sends a Bearer token.
Every route checks it.

The browser never sees the newsroom key. Authorization happens on the server, on every render.

FRONTEND

A public face and a working desk.

Next.js 16 + React 19 Server Components, Server Actions, partial prerender.

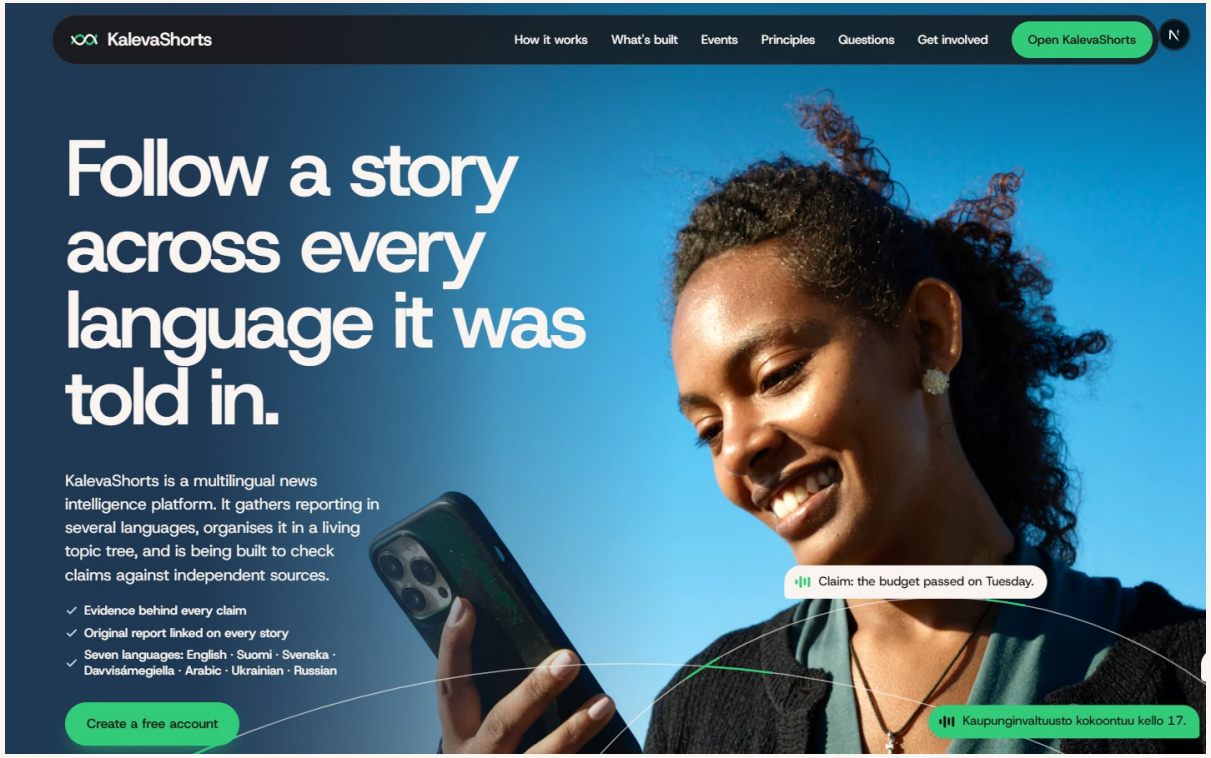
Clerk 7 Sign-in, session, roles; authorization checked server-side on every render.

GSAP 3 + Lenis Scroll scenes, SplitText, reduced-motion safe, clean teardown.

One design language Tokens from the landing page carried into the dashboard.

Seven languages Reader, editor and community surfaces; Arabic is right-to-left.

/dashboard /editor /dashboard/community



Public landing page · live screenshot

The same tokens, type and restraint run from the first screen into the editor.

EDITOR

The story tree, not a feed.

22 topics

11 branch links

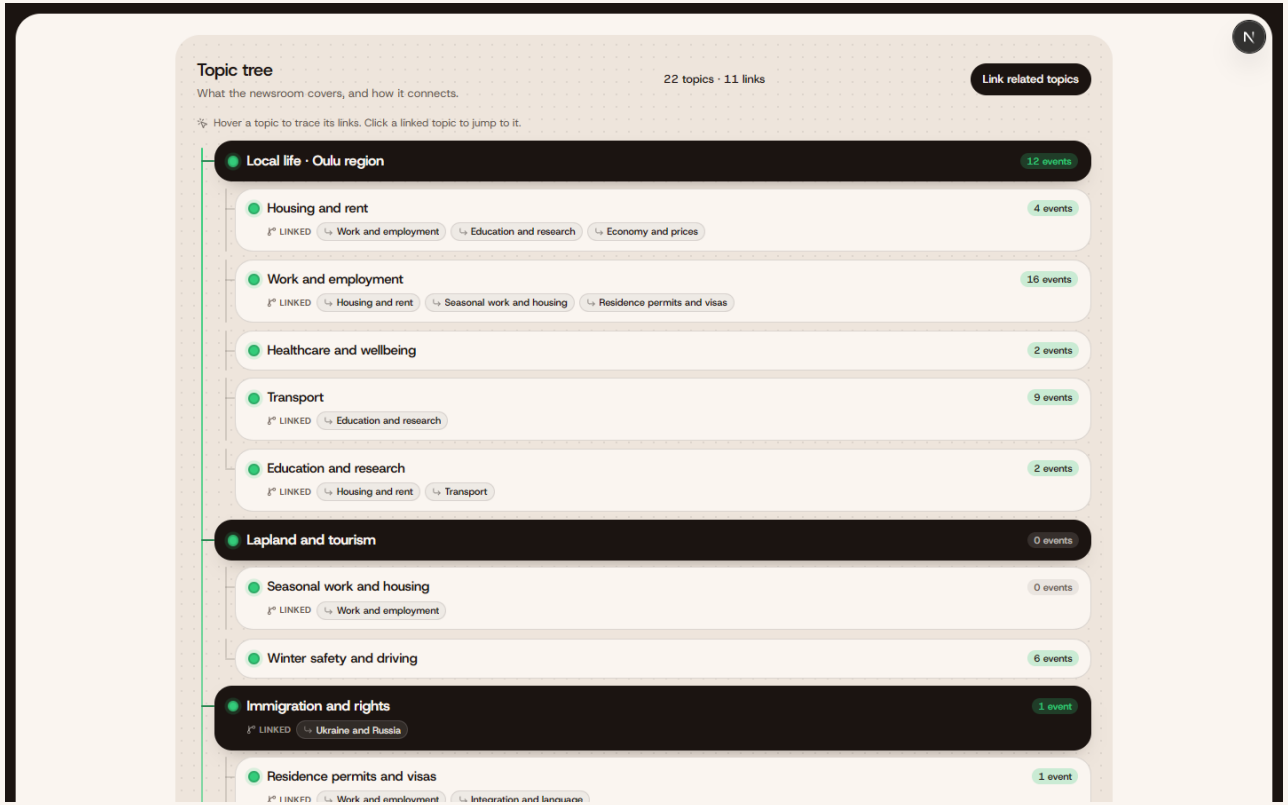
45 live events in the last 7 days

Events, not articles Articles attach to one event; the same news from several outlets becomes one row.

Symmetric branches The Connector links related topics both ways; the tree is a graph, not a folder.

Trace and jump Hover a topic to trace its links; click a linked topic to jump to it.

Drawn, then settled GSAP grows the trunk and branches, then hands the surface back.



Live editor data · real topic counts

A topic tree is how a newsroom explains its own judgement: what it covers, what it links, what it leaves empty.

BACKEND

The quiet engine.

01	**FastAPI /api/v1**	Async Motor, Pydantic v2, typed errors, OpenAPI.
02	**Clerk resource server**	RS256 verification, roles, lazy profile sync.
03	**Ingestion**	RSS, trafilatura, language detection, URL dedupe.
04	**Per-language store**	articles_en … articles_ru; one collection per edition.
05	**Related news**	Event, topic, branch, meaning, popularity, freshness.

RELATED NEWS, SCORED

score =

6	same event, another outlet
3	shared topic
1.5	branch topic
4 ×	embedding cosine
+	popularity × freshness

Every result carries its reasons.

76 backend tests · Ruff + format CI · no live services needed.

The browser asks for a language; the server returns one collection. No client-side data joins.

Agents do the work. People hold the gate.



9	newsroom roles	**LangGraph**	checkpointed workflow	**Every step**	audited with a reason
Bounded	investigation + revision	**Guarded**	untrusted text, prompt injection	**Human-only**	Sámi text + the final publish

Only a named person can publish. The agents can propose; they can never approve.

Trust is a workflow, not a banner.

01	**Named approvals**	Editor-in-chief, language reviewer, community reviewer, subject expert. A name, not a role label.
02	**Risk tiers**	Permits, health, work and legal rights need extra reviewers by policy.
03	**Corrections**	Public, named and attached to the story; the JSN guidelines become code checks.
04	**Sourcing**	Official references labelled; Kaleva metadata only; no paywall bypass.

Agents propose. Code validates. People decide.

REACH

Seven languages, one edition each.

English	46
Suomi	46
Svenska	46
العربية	46
Davvisámegiella	46
Українська	46
Русский	46

46 real stories

322 editions

45 events

5 Oulu sources

22 topics

10 communities

Live data in the current databases: the last seven days of real Oulu reporting, each story in all seven editions.

TODAY

The demo is the product.

READER

/dashboard

For-you feed in the chosen language

Related stories with reasons

Save, history, follow sources

EDITOR

/editor

Topic tree and source health

Run a cycle; review queue

Named approvals and corrections

COMMUNITY

/dashboard/community

Follow communities and official sources

Ask, tip, share local knowledge

Trusted voices and moderation

46 real Oulu stories · 322 editions · 45 events

last 7 days · every story in 7 languages

One platform, three systems, live data.

KalevaShorts

Local news that reaches everyone.

Ask us for a demo. Bring a story in any of the seven languages.

FRONTEND / BACKEND / AGENTIC NEWSROOM

Next.js 16 · FastAPI · LangGraph · Gemini · MongoDB

Kaleva Media Hackathon

Oulu · 2026